



PUAH YIDA

ENGINEERING PRODUCT DEVELOPMENT (EPD) STUDENT

CONTACT

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SKILLS

- Project Management
- Leadership
- Time Management
- Effective Communication
- Critical Thinking
- CAD: Fusion 360
- Prototyping: 3D Printing, Physical Testing, Iteration
- Programming: Python, MATLAB, Machine Learning

LANGUAGES

- English (Fluent)
- Chinese (Fluent)

EDUCATION

Bachelor of Engineering (Engineering Product Development) (2024-Present)

Singapore University of Technology
and Design (SUTD)

cGPA: 4.59/5.00

GCE Advanced Level (2020-2021)

Nanyang Junior College

Rank Points: 87.5/90

GCE Ordinary Level (2016-2019)

Catholic High School

LIR5 Raw Score: 7

ABOUT ME

I am a Term 5 junior and Global Merit Scholar at the Singapore University of Technology and Design (SUTD), with a strong interest in mechanical engineering and product design. I enjoy developing practical, user-centred solutions through CAD, prototyping, testing, and iteration.

I am especially interested in using modern design tools, including AI-assisted CAD and machine learning workflows, to improve the way products are designed and developed. Through my time at SUTD, I hope to deepen my mechanical engineering knowledge while building products that are meaningful, reliable, and useful in real-world applications.

AWARDS

- SUTD Honours List (Freshmore Terms), AY 2025 (2025)
- Sustainability Practice Prize (03.007 Design Thinking and Innovation) (2025)
- Certificate of Achievement (DIVE Global Exploration Opportunities programme) (2025)
- NYJC Academic Excellence Award (2022)
- NYJC Most Improved Student (2021)
- NYJC CCA Team Award - Chinese Orchestra (2021)
- Singapore Youth Festival Distinction - Chinese Orchestra (2021)

INVOLVEMENT

- **Vice-President of SUTD Climbers Club**
 - Oversee training and administrative tasks
 - Conduct Introduction Session for beginner climbers and new students
 - Organiser of Climbnival 2025
 - Open House 2025
 - DSUTD 2025
 - StuOrg Showcase 2025
 - Represented school in bouldering competitions (Gravical, ISBL, RISE, BoulderActive)
 - Organiser of Climbnival 2026
- **Member of Korean Language Enrichment Club (KLEC)**
- **Orientation Group Leader for ORI 2025**

WORK EXPERIENCE

- **Retail Associate @ The Cocoa Trees (Oct 2025 - Jan 2026)**
 - Assisted with product recommendation, enquiries and managed daily store operations including cashiering, restocking and product display at Changi Airport Terminal 3 North Outlet
- **Crew Staff @ Climb Central (Feb 2024 - Sep 2024)**
 - Supported daily climbing gym operations including customer check-ins, equipment rental, safety briefings, safety checks.

MAIN PROJECTS

HERMES – Automated Black Soldier Fly Neonate Counting & Singulation System SUTD, 30.007 Engineering Design Innovation • Worked with an 8-person engineering team through three formal design reviews (SRR, SDR, PDR), and contributed to system architecture decisions and ensuring functional integration of three subsystems (funnel, ramp, and camera) into a single working prototype • Designed and implemented an event-based computer vision counting algorithm (Python, OpenMV) using blob detection, track association, and coincidence-loss modelling, achieving 98.6% counting accuracy against hand-counted ground truth • CAD-designed the camera subsystem enclosure housing the event camera and LED lightbox, validated through live-neonate testing with 100% survival through the full system	2026
InVest – Smart Navigation Vest for Visually Impaired SUTD, 3D Design Project • Designed and prototyped an embedded assistive system integrating an ESP32-CAM with a computer vision pipeline for bus number recognition. • Implemented real-time image acquisition and wireless transmission for OCR-based decision-making • Developed a haptic feedback interface using vibration motors to provide directional navigation and bus arrival alerts. • Applied multi-disciplinary concepts from Structure & Materials, Computer-Aided Design & Digital Engineering, and Circuits & Electronics to enable system integration and performance optimisation	2026
Kaggle Machine Learning Challenge SUTD, 30.100 Computer & Data-Driven Engineering • Ranked 1st in cohort in a multi-task Kaggle competition involving three predictive modelling tasks. • Built end-to-end ML pipelines with feature engineering on structured and time-series data, and applied ensemble models (XGBoost, LightGBM, CatBoost) with cross-validation to ensure robustness. • Utilised LLM tools to accelerate iteration through code optimisation, feature ideation, and debugging.	2026
Solar-Powered Doorbell SUTD, 10.023 Designing Energy Systems • Reverse engineered a battery powered doorbell and modified it to be completely solar powered from scratch through precise power and energy analysis of the device to find the appropriate solar panel and battery requirements for the modification.	2025
AURAPLANE SUTD, 03.007 Design Thinking and Innovation • Together with group, designed a modular and multifunctional partition for both group and solo diners that optimises canteen seating availability and deters bird pest. • Design was awarded the Sustainability Practice Prize and attained an A for the module.	2025
Variable Inductor & FM Transmitter System SUTD, 10.017 Technological World • Designed and fabricated a coil-based variable inductor and integrated it with a custom LC circuit to build an FM transmitter • Applied electromagnetic theory to achieve target resonant frequencies and validated system performance through LCR measurements and software-defined radio signal reception.	2025
Mechanical Energy Powered Toy Car SUTD, 10.015 Physical World • Designed and prototyped a rubber band-powered vehicle using CAD and 3D printing. • Applied principles of energy storage, transmission, and friction to optimise performance through systematic testing of rubber band properties, wheel configurations, and drivetrain mechanisms under constrained design requirements	2024